

Module 4 Creating a Database and Tables using MariaDB

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Lab: Creating a Database and Tables using MariaDB

Objectives:

1. Create a MariaDB database.
2. Design and create a table named "Students" with columns: `StudentID`, `Firstname`, `Lastname`.
3. Design and create another table named "Courses" with columns: `CourseID`, `StudentID`, `CourseTitle`.
4. Establish a foreign key constraint on the "Courses" table referencing the "Students" table.

Instructions:

Step 1: Install MariaDB

- If not already installed, follow the instructions in the previous lab for installing MariaDB on your Windows or Mac PC.

Step 2: Launch MariaDB and Connect

- Launch the MariaDB command-line client and connect to the server using the appropriate command.

```
mysql -u root -p
```

Step 3: Create a Database

- Create a new database named "University" (or any preferred name).

```
CREATE DATABASE University;
```

Step 4: Use the Database

- Switch to the newly created database.

```
USE University;
```

Step 5: Create the "Students" Table

- Design and execute the SQL query to create the "Students" table.

```
CREATE TABLE Students ( StudentID INT PRIMARY KEY, Firstname VARCHAR(50), Lastname VARCHAR(50) );
```

Step 6: Create the "Courses" Table

- Design and execute the SQL query to create the "Courses" table.

```
CREATE TABLE Courses ( CourseID INT PRIMARY KEY, StudentID INT, CourseTitle VARCHAR(100), FOREIGN KEY (StudentID) REFERENCES Students(StudentID) );
```

Step 7: Verify Tables

- Confirm that the tables were created successfully.

```
SHOW TABLES;
```

Step 8: View Table Structure

- Optionally, view the structure of the "Students" and "Courses" tables.

```
DESC Students; DESC Courses;
```

Step 9: Exit MariaDB

- Exit the MariaDB command-line client.

```
EXIT;
```

Submission Requirements:

1. Save the SQL queries used to create the "Students" and "Courses" tables in a text file.
2. Include a screenshot showing the successful creation of the tables using the `SHOW TABLES` command.
3. Provide a brief explanation or comments in the submission document describing the purpose of each table and the foreign key relationship.

Note to Students: Ensure that your submission includes all the required components, and follow the documentation guidelines. Indicate any deviations from the standard instructions and provide explanations if certain steps were not executed successfully.

Grading Rubric: Creating a Database and Tables using MariaDB

Overall Score: 50 Points

Criteria:

1. Database Creation (10 points):

- Successfully created a new database named "University" or an alternative valid name.

2. Table Creation - Students (15 points):

- Executed correct SQL query to create the "Students" table.
- Verified the creation using the `SHOW TABLES` command.

3. Table Creation - Courses (15 points):

- Executed correct SQL query to create the "Courses" table.
- Verified the creation using the `SHOW TABLES` command.

4. Foreign Key Constraint (5 points):

- Established a foreign key constraint on the "Courses" table referencing the "Students" table.

5. SQL Query Submission (5 points):

- Submitted a text file containing the SQL queries used for creating the tables.

6. Screenshots (2 x 5 points):

- Submitted two screenshots:
 - Screenshot 1: Showing the successful creation of tables using the `SHOW TABLES` command.
 - Screenshot 2: Any relevant command output or execution confirming the foreign key constraint.

7. Documentation and Comments (5 points):

- Provided a brief explanation or comments in the submission document describing the purpose of each table and the foreign key relationship.

Notes:

- Ensure that SQL queries are syntactically correct and follow standard conventions.
- Screenshots should be clear and properly labeled.
- Documentation should be concise and address the purpose of each table and the foreign key relationship.

Points 50

Submitting a file upload

Due	For	Available from	Until
Oct 4	Everyone	-	Oct 22 at 11:59pm